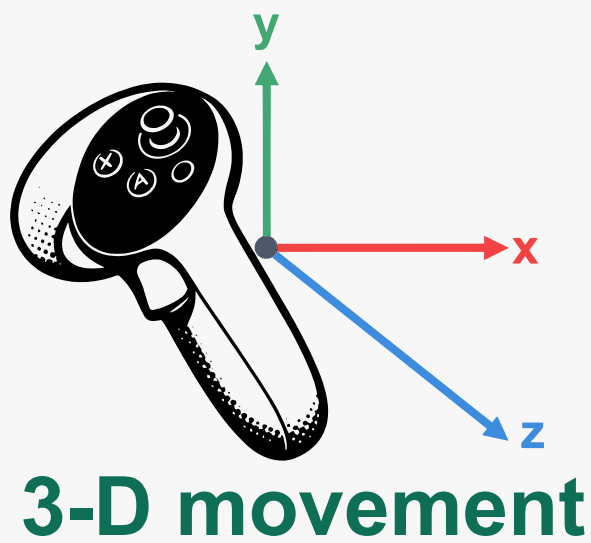




1 Capture 3-D belly movement

Abdomen-mounted controller pose read each frame.



1-D signal

2 Filter to 1-D breathing signal

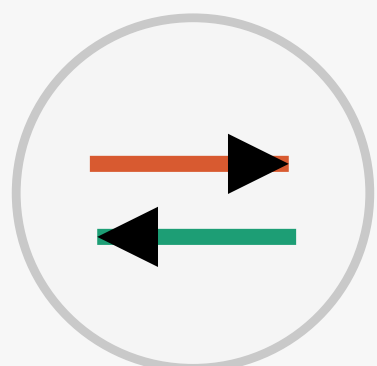
Projected onto calibrated axis. Range filter rejects non-respiratory motion. Dual smoothing yields 1-D trace.



Phase detection

3 Detect breath phase

Positive slope = **inhale**, negative = exhale, near-zero = hold. Signal reduced to three categorical states.



Phase mapping

4 Map phase to radius dynamic

Inhale drives expansion, exhale drives contraction, hold maintains radius.



Sphere radius

5 Integrate at fixed rate

Velocity set so full traversal takes the guided phase duration (here, 4 s).